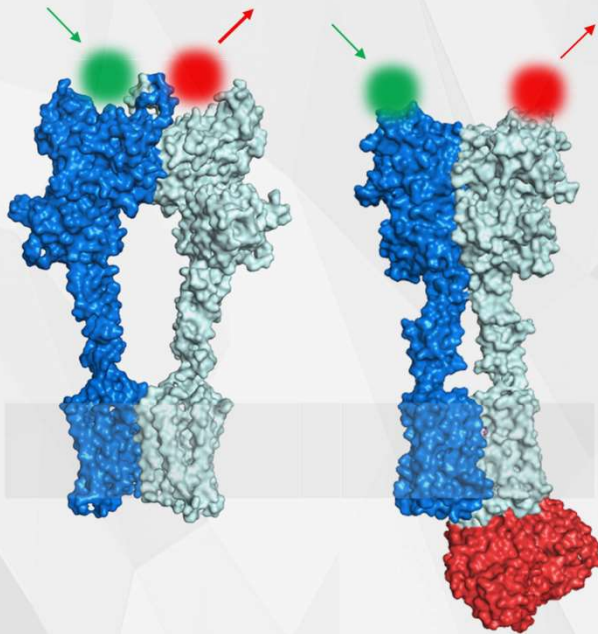


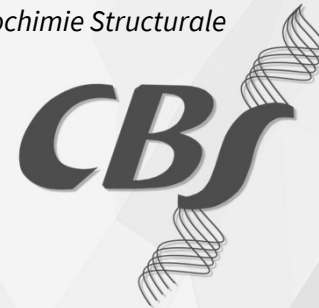
INTRODUCTION TO SINGLE MOLECULE FRET



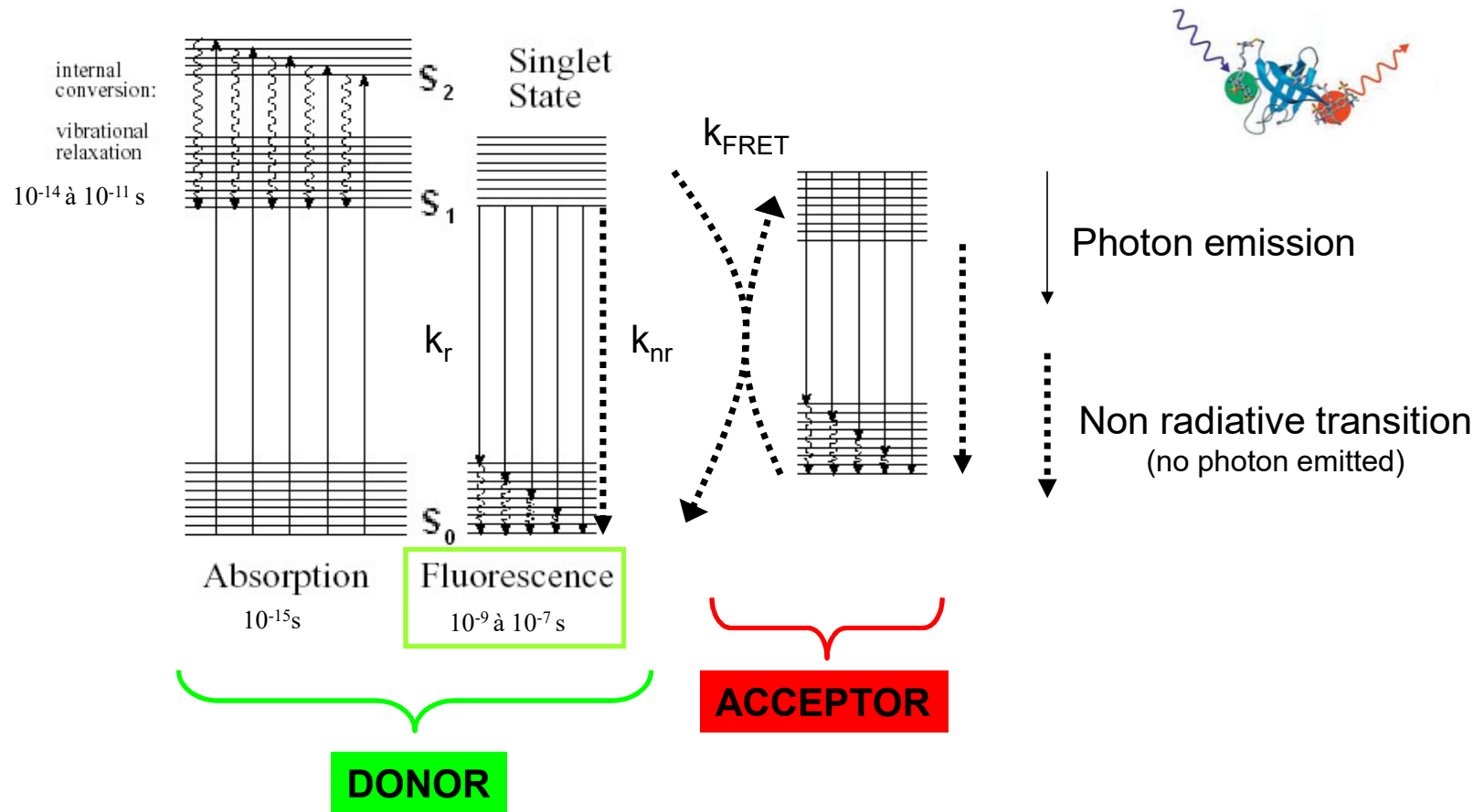
Emmanuel MARGEAT

Team Integrative Biophysics of Membranes

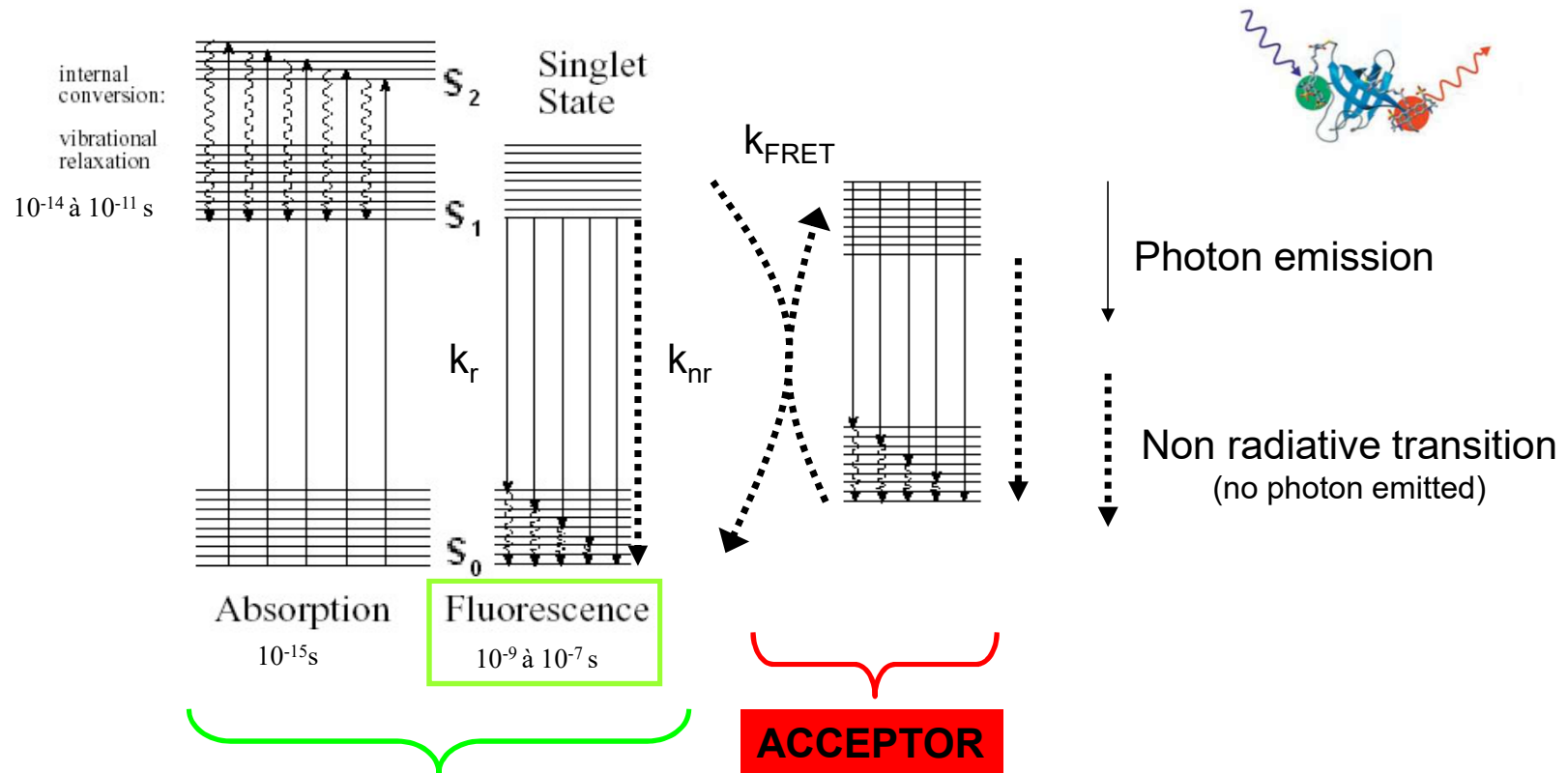
*Centre de Biochimie Structurale
Montpellier*



Simplified Jablonski diagram in the presence of FRET



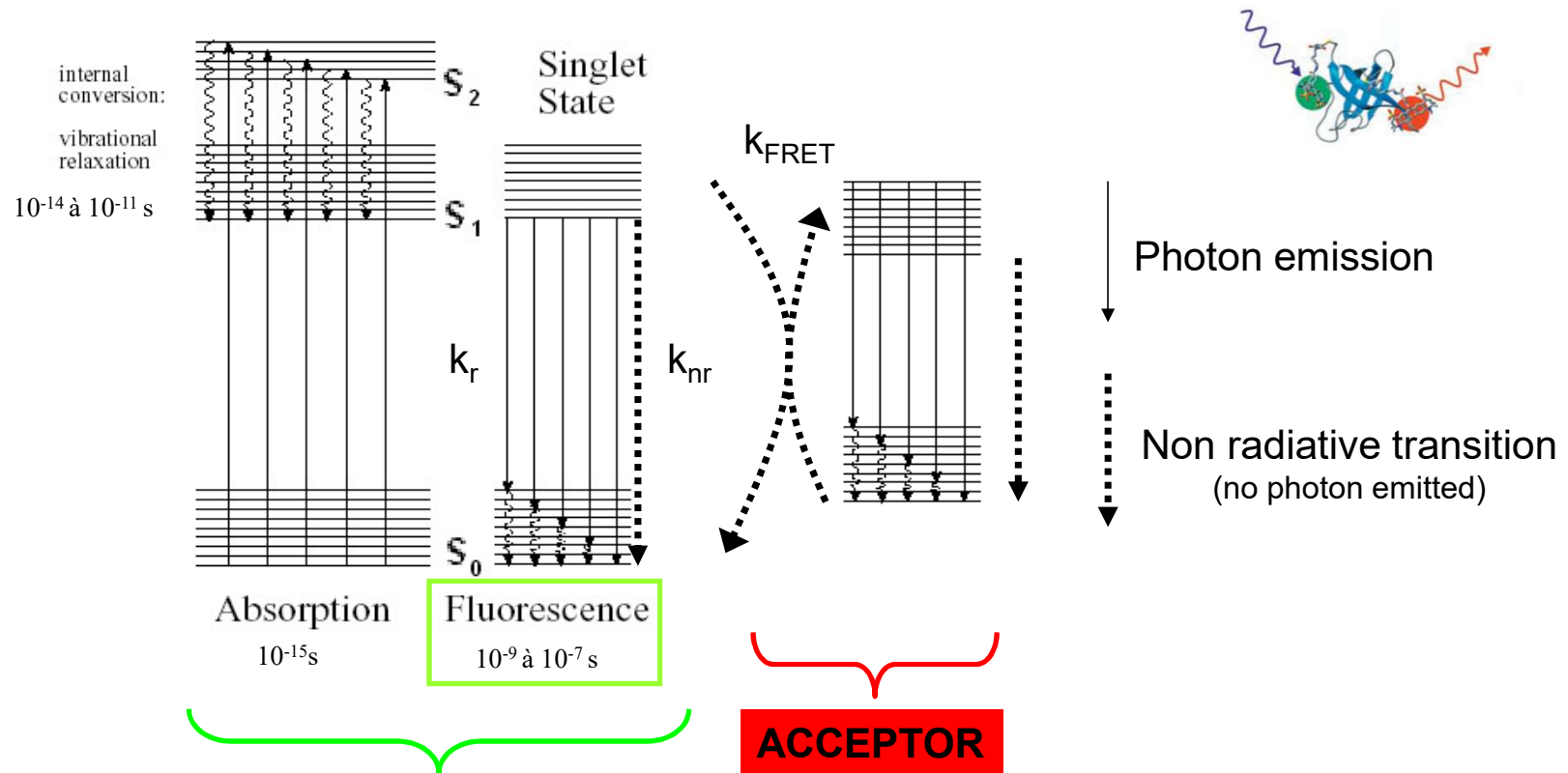
Simplified Jablonski diagram in the presence of FRET



$$\tau_D = \frac{1}{k_r + k_{nr}}$$

$$\tau_{D,A} = \frac{1}{k_r + k_{nr} + k_{FRET}}$$

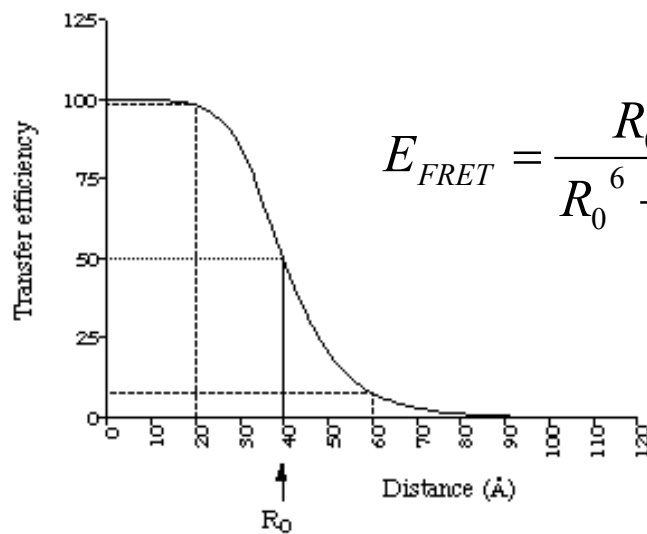
Simplified Jablonski diagram in the presence of FRET



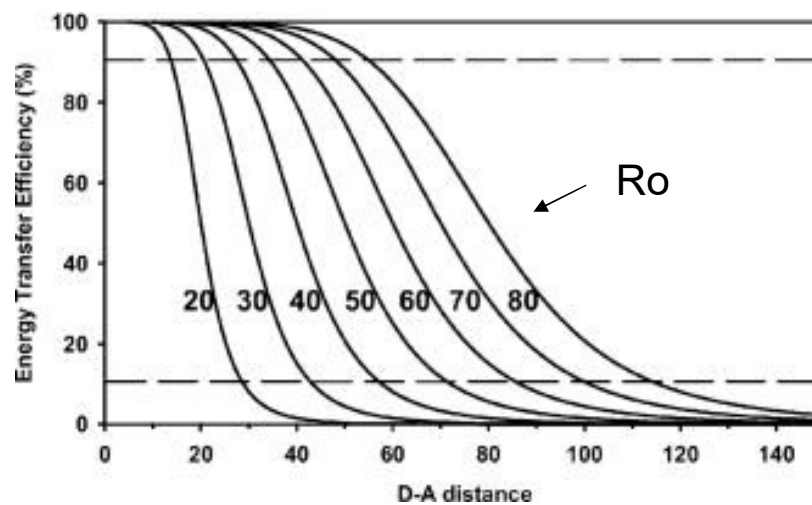
$$\tau_D = \frac{1}{k_r + k_{nr}}$$

$$\tau_{D,A} = \frac{1}{k_r + k_{nr} + k_{FRET}}$$

$$E = \frac{k_{FRET}}{k_r + k_{nr} + k_{FRET}} = 1 - \frac{\tau_{D,A}}{\tau_D}$$



$$E_{FRET} = \frac{R_0^6}{R_0^6 + R^6} = \frac{1}{1 + \left(\frac{R}{R_0}\right)^6}$$



→ FRET is a quantitative measurement of R between 0.5 and $1.5 R_0$

→ FRET is very sensitive to distance changes between 0.8 and $1.2 R_0$

FRET : Calculating R_0

$$R_0 = 9780 \cdot \sqrt[6]{\Phi_D \cdot \kappa^2 \cdot n^{-4} \cdot J(\nu)} \quad (\text{en } \text{\AA})$$

To obtain a distance, it is necessary to measure or estimate several factors

→ Φ_D is the quantum yield of the donor in the absence of the acceptor. This has to be measured relative to a reference standard (fluorescein, rhodamin, ..)

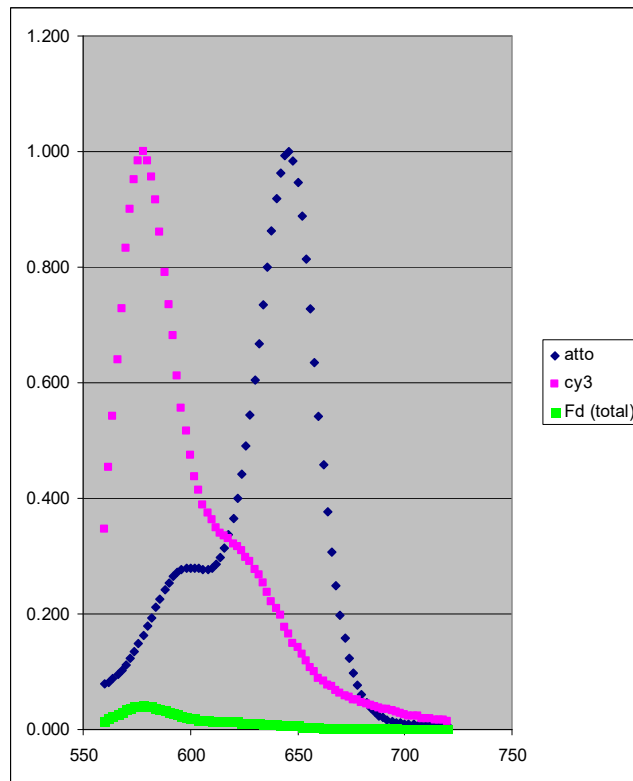
→ n = refractive index of the medium. For biological macromolecules, it is estimated to be : $n=1.4$

→ $J(\nu)$: spectral overlap, to calculate

→ κ^2 : orientation factor

FRET : the Spectral overlap integral J

$$J(\nu) = \int_0^{\infty} \left(\frac{\varepsilon^A(\nu) \cdot \phi(\nu)}{\nu^4} \right) d\nu$$



$\varepsilon^A(\nu) =$ Normalized excitation spectrum of A x $\varepsilon_{\max}^A$

$\phi(\nu)$ = Normalized fluorescence spectral shape function
 = Normalized emission spectrum of D, whose integral is normalized to 1

$$\nu = \lambda^{-1} \text{ (cm}^{-1}\text{)}$$

For each wavelength, calculate $\frac{\varepsilon^A(\nu) \cdot \phi(\nu)}{\nu^4}$

And sum over all wavelengths

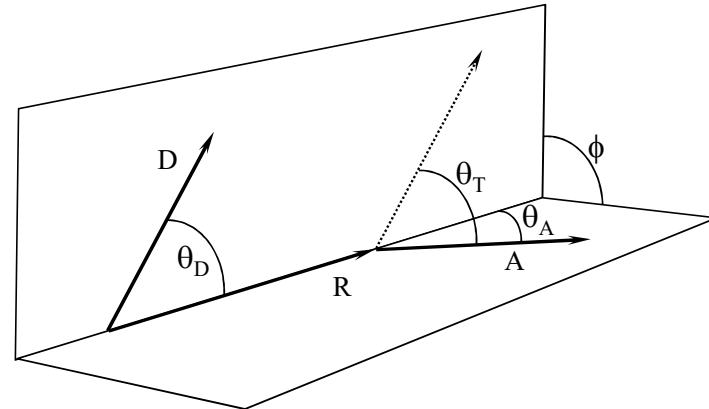
Here (cy3 / Atto647N) : $J = 6.0972 \cdot 10^{-13}$

FRET

The orientation factor κ^2

$$\kappa^2 = (\cos \theta_T - 3 \cos \theta_D \cos \theta_A)^2$$

where θ_T is the angle between the dipole moments of D and A, such as



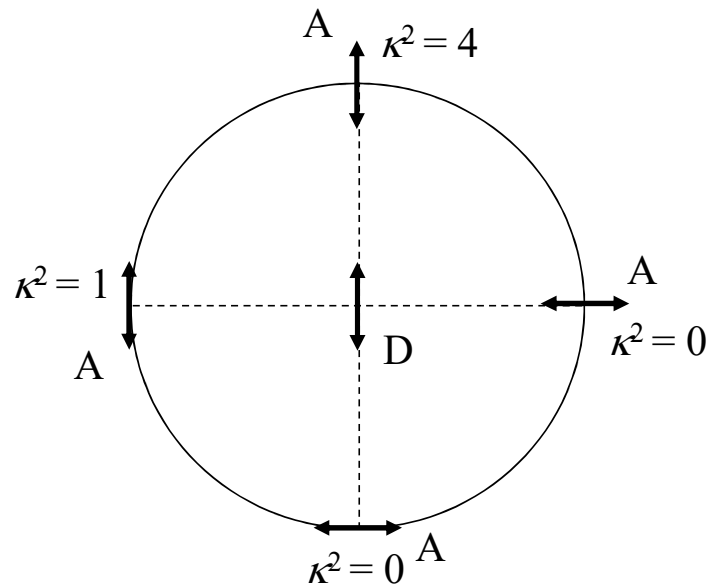
$$\cos \theta_T = \sin \theta_D \sin \theta_A \cos \phi + \cos \theta_D \cos \theta_A$$

where θ_D , θ_A are the angles between the separation vector R , and the dipole moments of D and A , respectively, and ϕ is the angle between the planes (D,R) and (A,R)

$$R_0 = 9,70.10^3 \left(\phi_D . J . \kappa^2 . n^{-4} \right)^{1/6}$$

FRET

The orientation factor κ^2

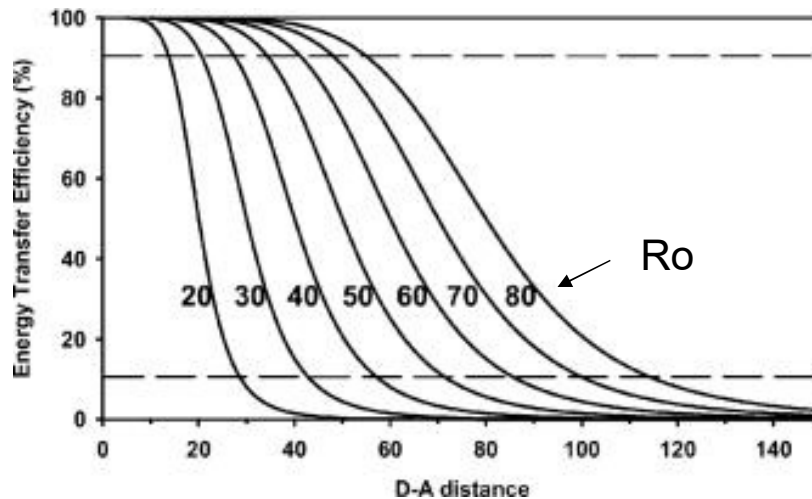


κ^2 varies from 0 to 4.

However, if the molecule reorients rapidly during the excited state lifetime of the dyes (dynamic averaging), we can show that $\kappa^2 \sim 2/3$

FRET

→ Molecular ruler that allows to measure distances within complexes and observe inter and intramolecular dynamics



Accessible distances depends on R₀

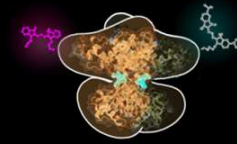
$$E_{FRET} = \frac{R_0^6}{R_0^6 + R^6} = \frac{1}{1 + \left(\frac{R}{R_0}\right)^6}$$

$$E = 1 - \frac{\tau_{D,A}}{\tau_D}$$

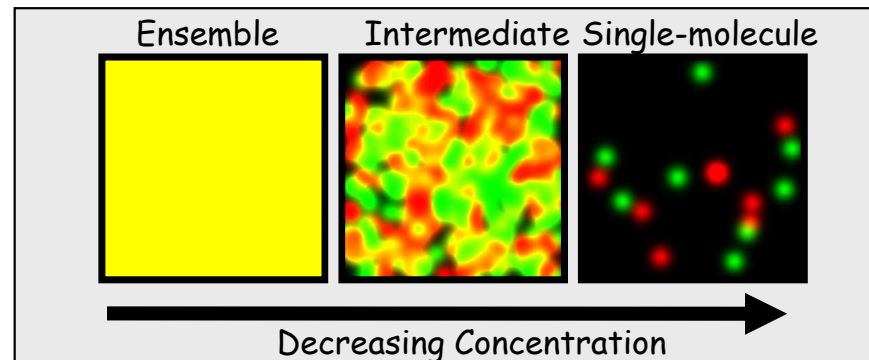
*E = calculated from intensities
of D and A
(requires correction factors)*

NB : FRET can exist between 2 identical fluorophores : this is called **homo-FRET**

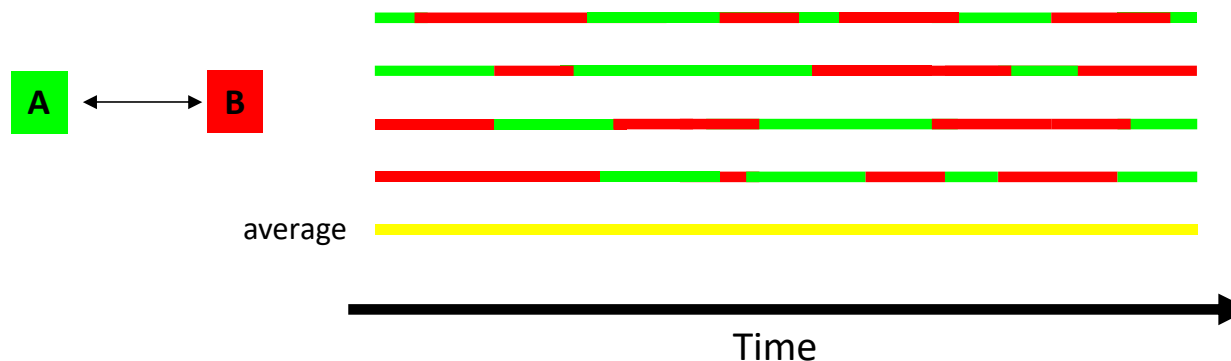
Why do we study single molecules ?



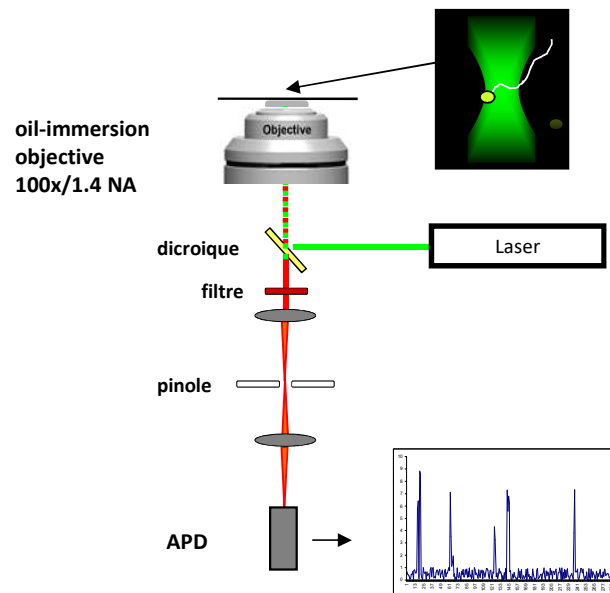
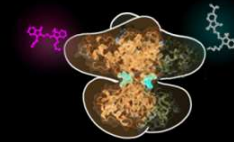
- Looking at ensemble of molecules generally leads to average values
- In particular, you cannot see **static heterogeneities**, i.e. the presence of subpopulations :



- Moreover, ensemble measurement do not allow to see the dynamics of unsynchronized molecules (**dynamic heterogeneities**)

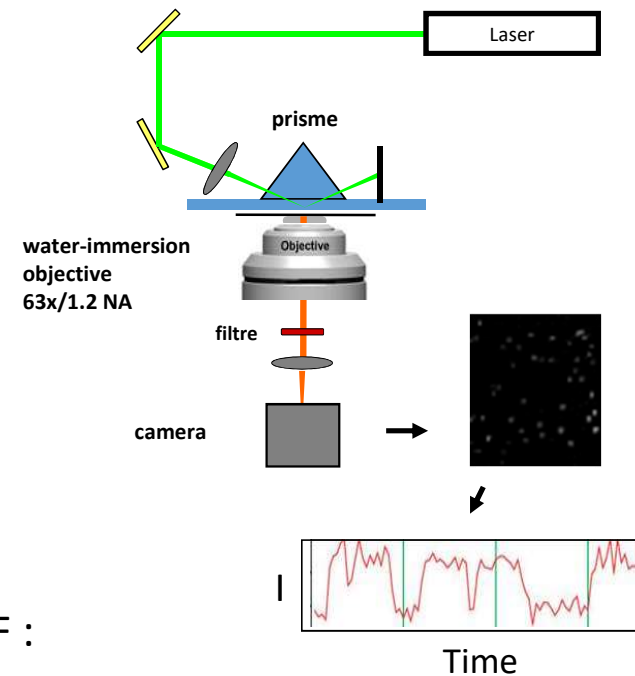


Confocal and TIRF microscopies



Confocal :

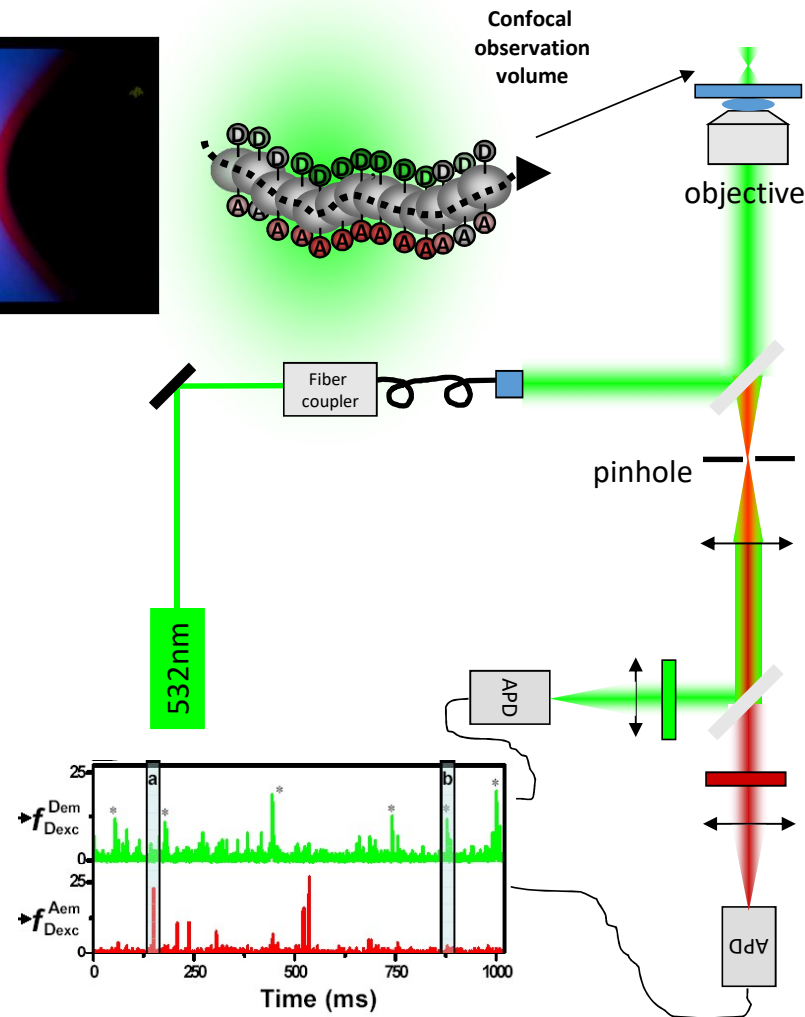
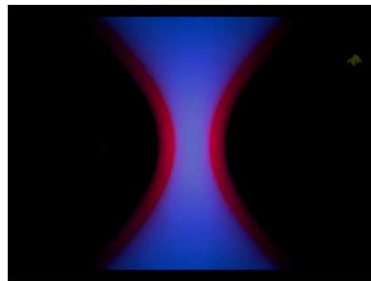
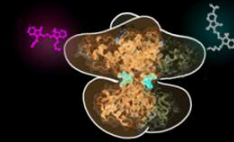
- immobilized or diffusing molecules in 3D
- 1 molecule at the time
- detection sub- populations
- High temporal resolution (μs -ns)



TIRF :

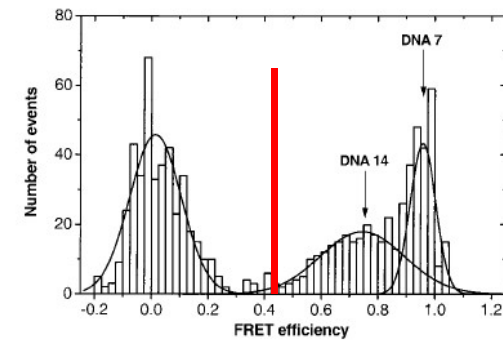
- immobilized or diffusing molecules in 2/3D
- dozens of molecules
- follow each molecules vs.time
- lower temporal resolution (ms)

smFRET on diffusing molecules



For each molecule :

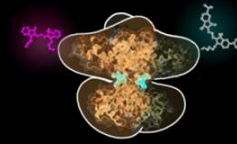
$$E_{PR} = \frac{F_{Aem}}{F_{Dem} + F_{Aem}}$$



→ resolve subpopulations

Deniz et al, *PNAS*, 1999

Single laser smFRET

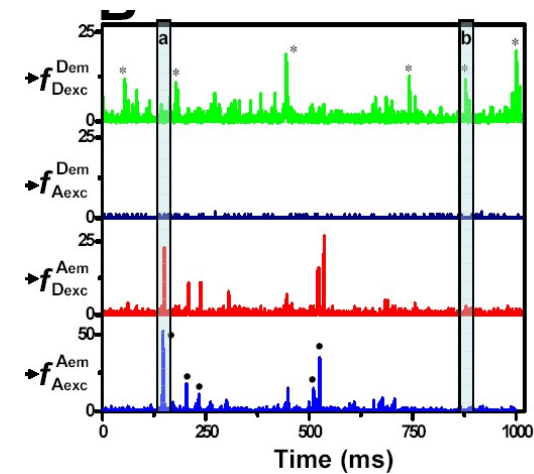
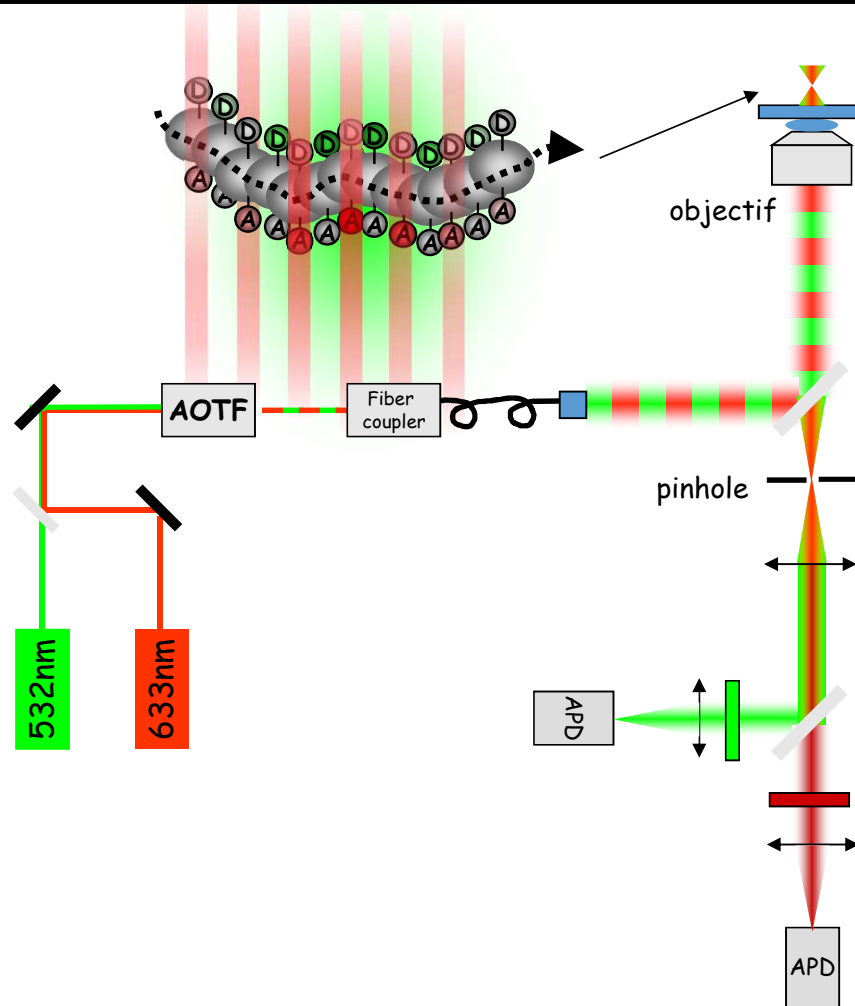
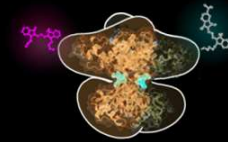


- + Not necessary to have a high yield of labeling, or to know it exactly
- + Ability to recover subpopulations
- + On immobilized molecules, spFRET traces = $f(\text{time})$
- Acceptor photophysics : blinking, bleaching : D-only species
- Sensitive to fluorescent contamination, appears as D-only
- Unable to recover the correction factors needed to extract quantitative FRET efficiency, and thus distances

→ Addition of a second laser to directly probe the emission of the acceptor

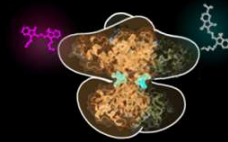
Alternating Laser EXcitation

Alternating laser excitation (ALEX) - smFRET



Kapanidis et al. *PNAS* 2004

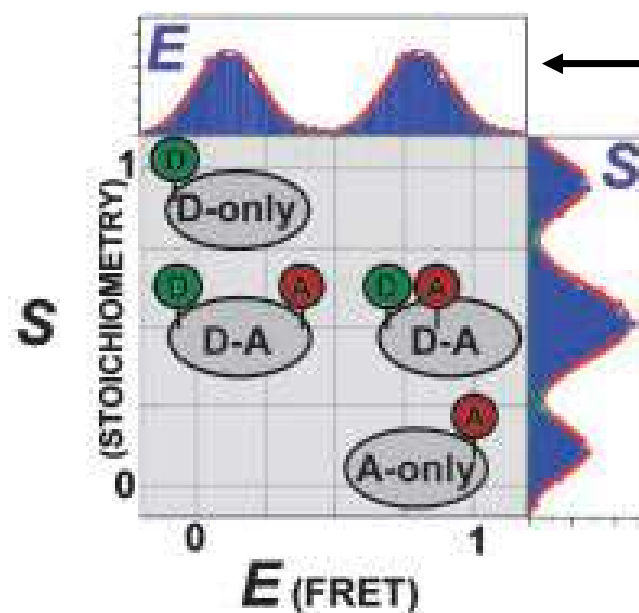
Identifying the species with ALEX



For each molécule

$$E_{PR} = \frac{F_{Dex, Aem}}{F_{Dex, Dem} + F_{Dex, Aem}}$$

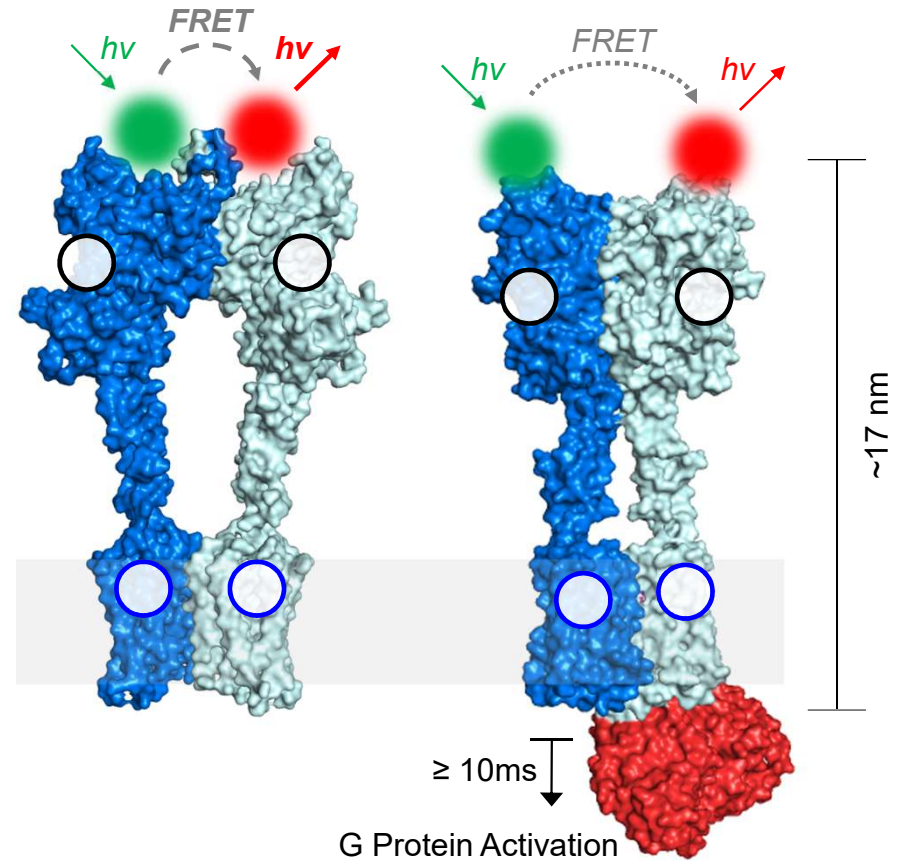
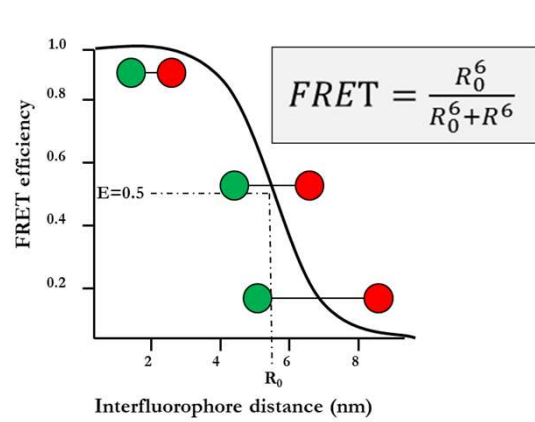
$$S = \frac{F_{Dex}}{F_{Dex} + F_{Aex}}$$



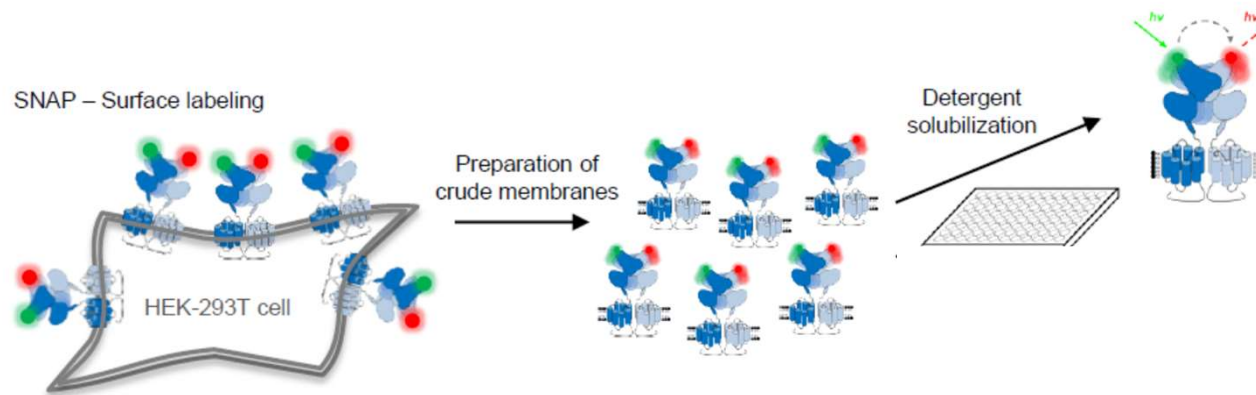
E_{PR} : distance between fluorophores

S : Presence of each fluorophores
→ Interaction
If D and A are on different molecules

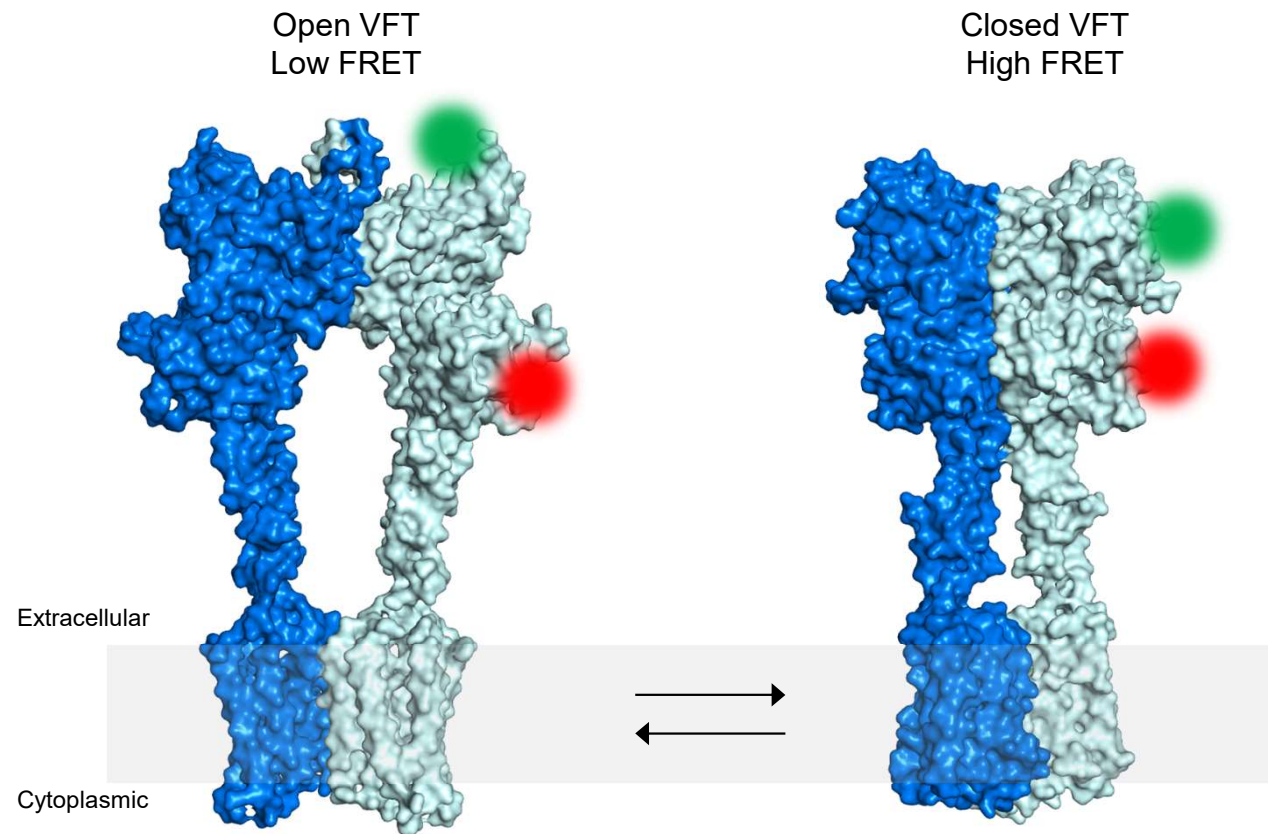
Structural dynamics of single metabotropic glutamate receptors



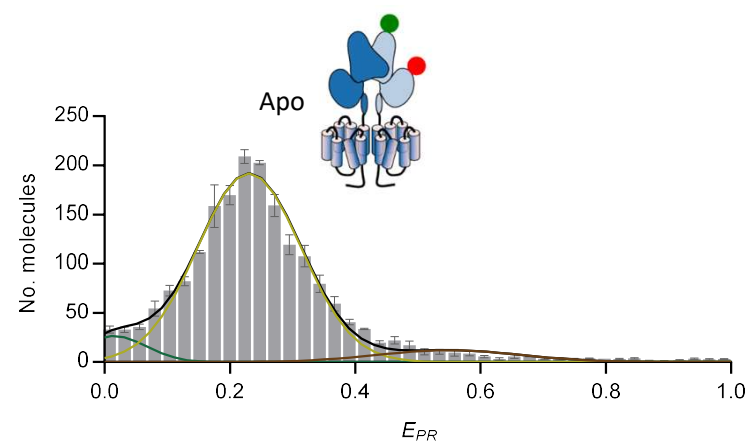
Structural dynamics of single metabotropic glutamate receptors



VFT Closure – Intrasubunit Sensor



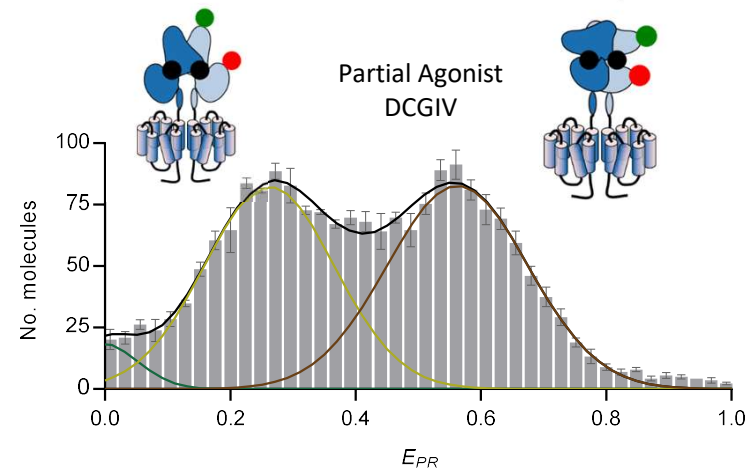
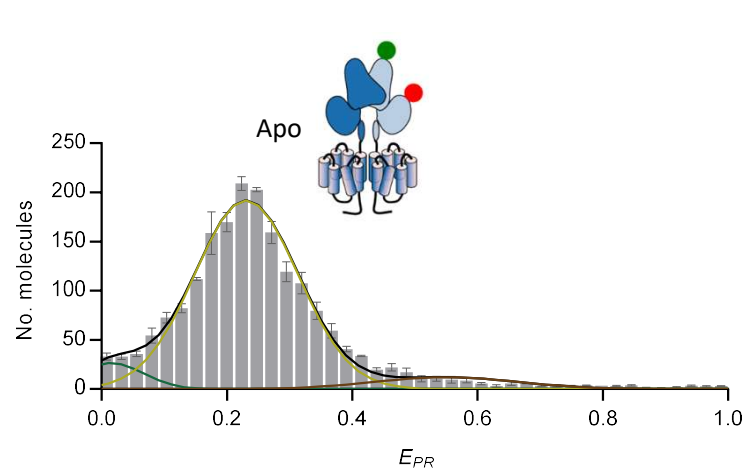
Agonist-dependent Effect on VFT Closure



Unpublished data

Quast et al., in preparation

Agonist-dependent Effect on VFT Closure



Unpublished data

Quast et al., in preparation

Agonist-dependent Effect on VFT Closure

